

USB4 2.0 ENGINEERING CHANGE NOTICE FORM

Title: Path Scheduler Accuracy Update
Applied to: USB4 Specification Version 2.0

Brief description of the functional changes:

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| <ul style="list-style-type: none">• Clarifying the accuracy requirement for each Path Scheduler with respect to their Weight assignment and for the USB3 Gen T Bundle Weight. |
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Benefits as a result of the changes:

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| <ol style="list-style-type: none">1. Alignment on spec requirements |
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An assessment of the impact to the existing revision and systems that currently conform to the USB specification:
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NA

An analysis of the hardware implications:
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NA

An analysis of the software implications:
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NA

An analysis of the compliance testing implications:
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NA

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Actual Change

(a). Section 5.3.3.1.1

From Text:

- The Path Scheduler shall schedule packets such that the number of bytes scheduled for each Path within the Priority Group is proportional to the weight assigned to that Path with 10% maximum deviation over a measurement period of 1 ms, assuming each Path in the Priority Group is either delivering packets throughout the 1 ms period or is idle throughout the 1 ms period, and that sufficient credits are available for each Path in the Priority Group

To Text:

- The Path Scheduler shall schedule packets such that the total number of packet header and payload bytes scheduled for each Path within the Priority Group is proportional to the weight assigned to that Path with 10% maximum deviation over a measurement period of 1 ms, assuming each Path in the Priority Group is either delivering packets throughout the 1 ms period or is idle throughout the 1 ms period, and that sufficient credits are available for each Path in the Priority Group

From Text:

A Path Scheduler in a Host Router that sets the *Gen T Bundle Weight Mode* bit to 1b operates with the following additional rules and changes for USB3 Gen T Paths:

- The shared weight assigned to all the USB3 Gen T Paths shall be determined by the *Bundle Weight* field in the USB3 Gen T Adapter Configuration Space of the USB3 Gen T Adapter connected to the USB4 Egress Port.
- When the *Bundle Weight* field changes, the Path Scheduler shall use the new shared weight.
- The Path Scheduler shall support weights in the range of 1-255 for the shared weight of the USB3 Gen T Paths.
- The Path Scheduler shall schedule packets such that the number of bytes scheduled for all the USB3 Gen T Paths together is proportional to the shared weight assigned in the *Bundle Weight* field with 10% maximum deviation over a measurement period of 1 ms.

To Text:

A Path Scheduler in a Host Router that sets the *Gen T Bundle Weight Mode* bit to 1b operates with the following additional rules and changes for USB3 Gen T Paths:

- The shared weight assigned to all the USB3 Gen T Paths shall be determined by the *Bundle Weight* field in the USB3 Gen T Adapter Configuration Space of the USB3 Gen T Adapter connected to the USB4 Egress Port.
- When the *Bundle Weight* field changes, the Path Scheduler shall use the new shared weight.
- The Path Scheduler shall support weights in the range of 1-255 for the shared weight of the USB3 Gen T Paths.
- The Path Scheduler shall schedule packets such that the total number of packet header and payload bytes scheduled for all the USB3 Gen T Paths together is proportional to the shared weight assigned in the *Bundle Weight* field with 10% maximum deviation over a measurement period of 1 ms.